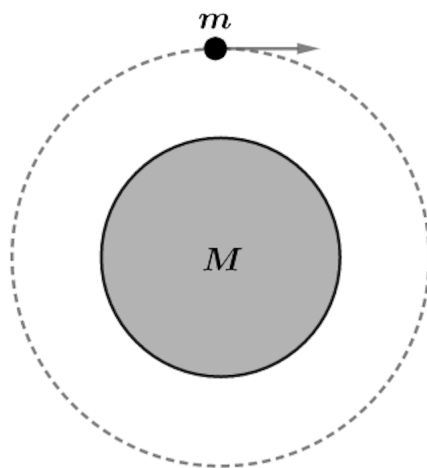


2018A F=ma Exam: Problem 15

Kevin S. Huang



The energy of the satellite is given by

$$E = K + U = \frac{1}{2}mv^2 - \frac{GMm}{R}$$

Since air resistance acts slowly over a long time, the trajectory will stay the same shape. For a circular orbit, gravity provides the centripetal acceleration:

$$\frac{GMm}{R^2} = \frac{mv^2}{R}$$

so

$$K = \frac{1}{2}mv^2 = \frac{GMm}{2R} = -\frac{U}{2}$$
$$E = K + U = \frac{U}{2}$$

Thus, if the satellite loses 1 J of total energy, it gains 1 J of kinetic energy and loses 2 J of potential energy so the answer is A.